
LatentGoalOps: A Benchmark for Latent Objective Adaptation Under Silent Shifts

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Latent objectives are common in real operational settings: stakeholders reveal con-
2 straints, preferences, and feedback over time, while the true utility function gov-
3 erning good decisions may remain implicit or change without announcement. We
4 introduce **LatentGoalOps**, a benchmark for evaluating latent objective inference
5 and adaptation under silent objective shifts. Agents observe a fixed public oper-
6 ating schema consisting of business state, stakeholder messages, decision ledgers,
7 delayed effects, and governance constraints, but never observe the hidden objec-
8 tive used by the deterministic grader. The benchmark contains three sequential
9 business-operations environments; our main analysis focuses on two week-long
10 anchor tasks, with a shorter planning task used as breadth evidence. Across the
11 anchor tasks, restricted-view leakage audits show no easy shortcut, models incur
12 a persistent penalty on silently shifted episodes, and the best current local model
13 remains roughly 0.20 below a myopic oracle that observes the active hidden objec-
14 tive but does not solve the full future trajectory. We further find that performance is
15 mechanism-sensitive: several sharp heldout drops are caused by structured-action
16 failures rather than latent-policy quality alone. For this reason, LatentGoalOps re-
17 ports strict score, native-only score, and interface reliability jointly. These results
18 position LatentGoalOps as a benchmark for studying hidden-objective adaptation
19 under a fixed public schema and deterministic grader, rather than as a measure of
20 general business-agent readiness.

1 Introduction

Many agent benchmarks make the objective explicit: the instruction is written down, the tool contract is visible, and success is defined against an external goal. Recent suites such as τ -bench and BrowserGym have pushed that setting toward longer-horizon, tool-using evaluation in realistic interfaces [Yao et al., 2024, Le Sellier De Chezelles et al., 2024]. Real operating environments, however, often require a different capability. Leadership priorities are not always fully stated, stakeholder messages are noisy, visible KPI movement is ambiguous, and the underlying objective can change without announcement.

This paper targets that gap. We study *latent objective inference and adaptation*: inferring an unstated reward structure from public evidence, acting coherently through a strict interface, and adapting when the hidden objective changes mid-episode. We instantiate this construct in **LatentGoalOps**, a synthetic but structured operating-world benchmark with persistent entities, delayed effects, deterministic grading, and silent objective shifts. The synthetic construction is intentional: it gives the benchmark hidden-objective ground truth while preserving a realistic operating surface for the agent.

The release contains three paper-facing sequential environments: *P1 Startup Week*, *P2 Incident Response Week*, and *P3 Quarterly Headcount Plan*. The paper centers P1 and P2 as anchor tasks; P3 serves only as shorter breadth evidence. Within that scope, the frozen evidence supports three claims: current local open-weight configurations retain a clear myopic-oracle gap, within-family behavior is heterogeneous rather than uniformly monotone, and some large heldout drops are interface-mediated rather than pure policy regressions.

Related work and positioning. Existing interactive benchmarks such as τ -bench and BrowserGym primarily evaluate explicit user goals in realistic interfaces, while structured-output and function-calling evaluations such as JSONSchemaBench and BFCL isolate interface control itself [Yao et al., 2024, Le Sellier De Chezelles et al., 2024, Geng et al., 2025, Patil et al., 2025]. LatentGoalOps instead hides the objective, introduces delayed effects and silent shifts, and requires strict structured actions under non-stationarity. Following recent benchmark-methodology work, we therefore report strict score, native-only score, and interface reliability together rather than a single leaderboard scalar [Authors, 2025, Reuel et al., 2024, Mazumder et al., 2023, Longjohn et al., 2024].

Contributions.

- We introduce LatentGoalOps, a benchmark suite for latent objective inference and adaptation under silent non-stationarity with delayed effects, hidden strategic objectives, deterministic grading, two week-long anchor environments, and one shorter breadth task.
- We release a runnable benchmark artifact with a frozen protocol, core and heldout splits, deterministic baselines, restricted-view leakage probes, generated paper assets, and reviewer-facing reproducibility documentation.
- We report a strengthened local benchmark matrix with 630 primary model episodes and 360 baseline episodes, plus a P1/P2 anchor-task robustness slice that isolates the benchmark’s strongest latent-adaptation settings.
- We show that the benchmark’s empirical story is mechanism-sensitive: some larger-model changes are native-policy movements, while some of the sharpest heldout regressions are interface-mediated drops.

Benchmark	Business ops	Hidden objective	Silent shift	Delayed effects	Strict schema
WebArena	No	No	No	No	No
WorkArena	Yes	No	No	No	No
BrowserGym	No	No	No	No	No
OSWorld	No	No	No	No	No
τ -bench	Yes	No	No	Limited	No
LatentGoalOps	Yes	Yes	Yes	Yes	Yes

Table 1: Qualitative construct comparison against nearby interactive-agent benchmarks [Zhou et al., 2023, Drouin et al., 2024, Le Sellier De Chezelles et al., 2024, Xie et al., 2024, Yao et al., 2024]. LatentGoalOps differs by combining hidden objective, silent shift, delayed effects, and a strict executable action shell.

2 Benchmark

2.1 Environment Loop

Each episode is a finite partially observed decision process. At step t , the agent receives a public observation o_t , emits a structured action a_t , and the environment advances to the next state. Public observations include inbox messages, KPI dashboards, dated decision ledgers, pending effects, realized effects, customer accounts, stakeholder personas, team state, market context, and visible governance constraints. The hidden objective g_t is never shown directly.

Formally, an episode is a trajectory

$$\tau = (o_0, a_0, o_1, a_1, \dots, o_T),$$

with hidden state x_t , public observation $o_t = \psi(x_t)$, and active hidden objective state g_t . The environment maintains a normalized four-channel metric vector $m_t \in [0, 1]^4$ over growth, retention, revenue, and efficiency, and an active latent weight vector $w_t \in \Delta^4$. The latent utility used by the grader is

$$U_t = \text{clip}_{[0,1]} \left(\sum_{c \in \mathcal{C}} m_{t,c} w_{t,c} \right), \quad \mathcal{C} = \{\text{growth, retention, revenue, efficiency}\}.$$

The action interface is intentionally strict. On the sequential tasks, a valid action is a JSON-serializable payload over task-specific fields such as `chosen_initiatives`, `messaging_action`, `pricing_change_pct`, `support_policy`, or `budget_allocations`. If the model fails to emit a valid structured action, the benchmark executes a fallback action and the trajectory still advances. In other words, interface failures are part of the task, not merely parser trivia. This follows the broader lesson from structured-generation and tool-use evaluation that output control is part of agent capability rather than an implementation footnote [Geng et al., 2025, Patil et al., 2025].

2.2 Three Sequential Environments and Silent Shifts

LatentGoalOps evaluates three sequential operating environments. The implementation preserves internal task ids for reproducibility, but the main paper uses the cleaner P1–P3 names throughout. Table 2 summarizes the benchmark environments and their roles in the manuscript.

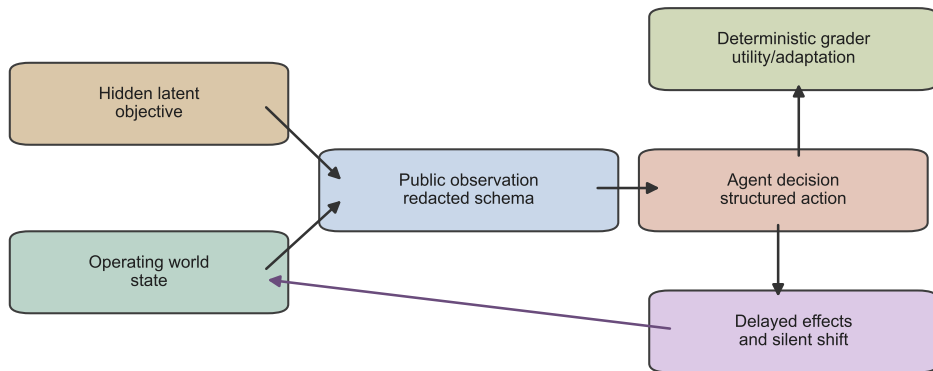
P1 Startup Week is a ten-step operating week with initiatives, renewals, and delayed business consequences. P2 Incident Response Week is a seven-step incident setting where trust repair, governance, recovery speed, and cost can conflict. P3 Quarterly Headcount Plan is a shorter staffing task used only for breadth. Heldout evaluation changes scenario families and delayed-effect realizations while preserving the same public schema and grading semantics. Silent shifts occur with probability 0.6, usually at steps $\{3, 4, 5\}$, and can alter latent weights as well as risk posture, planning horizon, segment focus, and governance strictness. The model is never told that the shift occurred.

2.3 Scoring and Reporting

For the primary sequential tasks, the episode score is an explicit weighted combination of final latent utility U , adaptation A , coherence C , and constraint satisfaction K :

$$S_{\text{task}} = \alpha_U U + \alpha_A A + \alpha_C C + \alpha_K K.$$

LatentGoalOps evaluates whether agents infer and adapt to hidden business objectives



The agent only sees public observations. The latent objective and delayed consequences shape future rewards.

Figure 1: LatentGoalOps exposes public operating state and delayed consequences while hiding the objective that determines utility. The agent must act through a strict structured-action interface; failures at that interface alter the trajectory itself.

Paper ID	Environment	Horizon	Paper role	Hidden shift
P1	Startup Week	10	anchor	Yes
P2	Incident Response Week	7	anchor	Yes
P3	Quarterly Headcount Plan	4	breadth	Yes

Table 2: LatentGoalOps benchmark environments. The main paper’s result-bearing analyses center the two week-long anchor tasks P1 and P2.

96 Task-specific coefficients and no-shift renormalization rules are fixed in the release protocol. This
 97 keeps the main paper focused on the measurement construct rather than coefficient bookkeeping.

98 For *Startup Week* and *Incident Response Week*, adaptation measures how quickly post-shift decisions
 99 improve under the active hidden objective and whether that recovery is sustained; P3 uses the same
 100 principle with a shorter horizon. We report overall strict score, native-only score, shifted versus
 101 unshifted score, and interface reliability. Release aggregates pool 10 seeds per task across P1–P3,
 102 while P1/P2 anchor slices isolate the strongest construct signal; aggregate views use 95% bootstrap
 103 confidence intervals.

104 3 Evaluation and Benchmark Validity

105 3.1 Baseline Separation

106 LatentGoalOps has meaningful score dynamic range on the frozen release aggregate over P1–P3.
 107 Random, visible-heuristic, and myopic-oracle baselines preserve the expected ordering on both
 108 splits, and the myopic-oracle gap remains substantial on both splits. The myopic oracle observes
 109 the active hidden objective and chooses the best immediate action, but does not solve the full future
 110 trajectory; we use it as a construct headroom reference rather than as a fair agent baseline or human
 111 proxy.

112 3.2 Leakage Audit

113 Restricted-view centroid probes test for shortcut access to the hidden objective. Table 4 shows
 114 that P1 and P2 stay near the four-way chance regime under both full and restricted views, with
 115 `explicit_goal_token_rate` equal to 0.0 throughout. P3 is less clean: its core restricted-view
 116 maximum reaches 0.35, which is why we treat it as breadth evidence rather than as a validity anchor.
 117 Each probe slice uses 20 train and 20 test examples, so this audit is a shortcut screen rather than a
 118 proof of impossibility.

Baseline	Split	Episodes	Mean score	95% CI
random	core	60	0.303	[0.286, 0.320]
visible heuristic	core	60	0.346	[0.329, 0.362]
myopic oracle	core	60	0.599	[0.575, 0.623]
random	heldout	60	0.310	[0.290, 0.330]
visible heuristic	heldout	60	0.371	[0.356, 0.388]
myopic oracle	heldout	60	0.598	[0.575, 0.623]

Table 3: Baseline separation on the frozen primary-task aggregate.

Task	Split	Full	Numeric	Sender	Order	Max restricted	Goal tokens
Startup Week	core	0.20	0.05	0.25	0.25	0.30	0.00
Startup Week	heldout	0.20	0.15	0.20	0.25	0.25	0.00
Incident Response Week	core	0.20	0.10	0.30	0.25	0.30	0.00
Incident Response Week	heldout	0.15	0.20	0.30	0.25	0.30	0.00
Quarterly Headcount Plan	core	0.25	0.25	0.25	0.25	0.35	0.00
Quarterly Headcount Plan	heldout	0.20	0.15	0.25	0.25	0.30	0.00

Table 4: Compact restricted-view leakage summary for paper-facing tasks. Chance is 0.25 for the four-way archetype probe. “Max restricted” is the maximum over non-full views in the frozen audit and is interpreted as a screening diagnostic rather than a proof of no shortcut.

Utility-only rescoring sharply compresses the myopic-oracle gap, which is the expected pattern if the benchmark is measuring latent-objective adaptation rather than only final-state utility.

4 Main Results

4.1 The Two Week-Long Anchor Tasks Carry the Main Result

Because P3 is shorter-horizon and less discriminative than the two week-long environments, Table 5 reports a P1/P2-only robustness slice computed directly from the frozen release as the paper’s main result view. On the anchor slice, the strongest local core mean is Qwen 27B at 0.425 and the strongest heldout mean is DeepSeek 8B at 0.431, while the myopic-oracle anchor means are 0.626 on core and 0.630 on heldout. The best local anchor results therefore still trail the oracle by about 0.20.

Taken together with the leakage audit and the persistent shifted-versus- unshifted gap, the P1/P2 anchor tasks make the intended construct visible: public trajectory evidence is informative but incomplete, so successful behavior requires latent-goal tracking rather than shortcut extraction.

4.2 The Full Release Aggregate Preserves the Same Story

The full P1–P3 aggregate preserves the same story: the strongest frozen local aggregates are 0.443 on core and 0.446 on heldout, still about 0.15 below the myopic-oracle baselines of 0.599 and 0.598.

4.3 Some of the Sharpest Heldout Drops Are Interface-Mediated

Across the frozen release, scaling is family-specific rather than monotone, but the most important mechanism is interface confounding rather than raw ranking alone. DeepSeek remains the cleanest native-policy signal: its three reported sizes stay at native rate 1.0 on both splits, so its gains and losses are best read as policy movement. Qwen 2B $\rightarrow$ 9B also changes at native rate 1.0, giving a clean example of non-monotone native-policy behavior. The sharp heldout failures, by contrast, are dominated by shell fragility.

The clearest examples are Gemma 26B and Qwen 27B. On the P1/P2 anchor slice, Gemma 26B heldout falls to 0.341 overall at native rate 0.70, while Qwen 27B heldout falls to 0.340 overall at native rate 0.25. On the full release aggregate, Qwen 27B heldout rises from 0.385 overall to 0.457 on the native subset, but that subset contains only 14 episodes and must be read diagnostically rather than as a leaderboard claim. Table 6 grounds that mechanism in exact stored outputs: DeepSeek

Model	Core P1/P2	Heldout P1/P2	Core native rate	Heldout native rate
Myopic oracle	0.626	0.630	1.00	1.00
DeepSeek 1.5B	0.407	0.407	1.00	1.00
DeepSeek 8B	0.408	0.431	1.00	1.00
DeepSeek 14B	0.420	0.405	1.00	1.00
Gemma 2B	0.401	0.412	1.00	1.00
Gemma 4B	0.408	0.416	1.00	1.00
Gemma 26B	0.392	0.341	0.65	0.70
GPT-OSS 20B	0.407	0.358	1.00	1.00
Qwen 2B	0.406	0.412	1.00	1.00
Qwen 9B	0.394	0.393	1.00	1.00
Qwen 27B	0.425	0.340	0.95	0.25

Table 5: P1/P2 robustness slice from the frozen release. The myopic-oracle row makes the remaining adaptation headroom explicit, while the weakest heldout cells coincide with shell-fragile models.

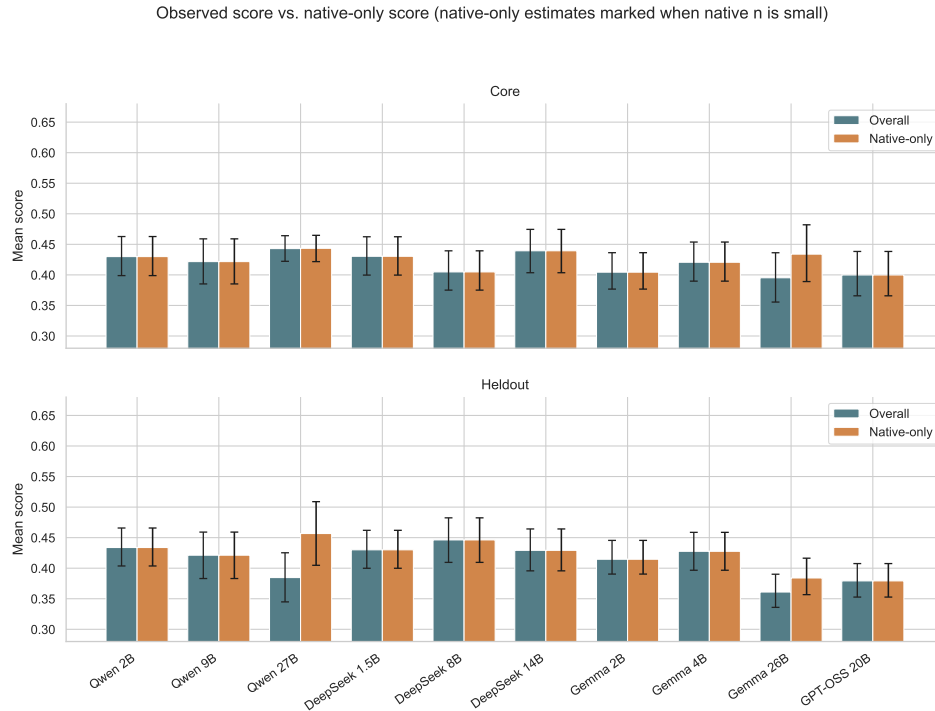


Figure 2: Observed score versus native-only score on the frozen release aggregate. Small-native-count estimates are diagnostic only, but the figure makes shell confounding visible instead of hiding it inside a single scalar mean.

147 stays incident- focused across the shift, whereas Qwen 27B hits an empty fallback and later returns
148 stale growth/revenue moves.

149 4.4 Silent Shifts Remain the Dominant Stressor

150 Silent objective shifts remain the primary benchmark stress test. Across the release matrix, shifted
151 episodes are materially harder than unshifted ones on both core and heldout, with a median unshifted-
152 minus-shifted gap of 0.172 in the release-wide diagnostics. This gap is not explained away by
153 shell failures: models with perfect native-action reliability, including all DeepSeek sizes, still pay a
154 substantial shift penalty. That is the benchmark construct we most want to isolate. Figure 3 makes
155 this visible at the aggregate level.

Run / step	Public clue	Selected output fields	Strict / reward
DeepSeek 14B, P2 day 2	Day 2; 3 pending effects	<code>initiative=ship_usage_pricing;</code> <code>message=retention_campaign; pricing=-0.1;</code> <code>support=incident_swarm</code>	strict; -0.0302
DeepSeek 14B, P2 day 8	Day 8; 0 pending effects	<code>initiative=refactor_incident_tooling;</code> <code>message=retention_campaign; pricing=0.0;</code> <code>support=incident_swarm</code>	strict; 0.1427
Qwen 27B, P1 day 5	Day 5; 0 pending effects	<code>{}</code>	non-strict; -0.0216
Qwen 27B, P1 day 9	Day 9; 3 pending effects	<code>initiative=launch_referral_loop;</code> <code>message=growth_push; pricing=0.0;</code> <code>support=balanced_triage</code>	strict; -0.2013

Table 6: Exact selected fields copied from stored action payloads and episode logs in the frozen case studies. These are action outputs and scalar episode diagnostics, not chain-of-thought traces.

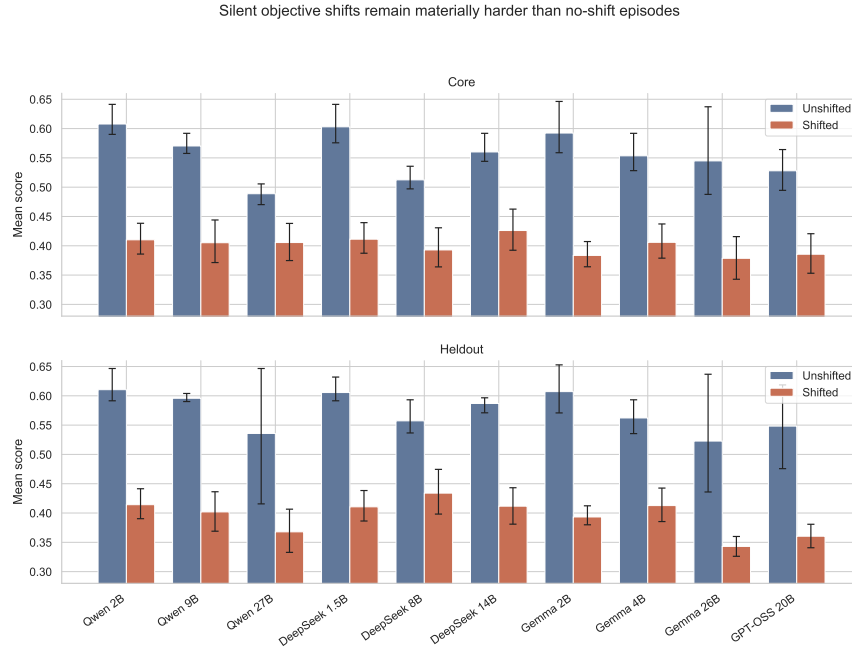


Figure 3: Shifted episodes remain materially harder than no-shift episodes on the frozen primary-task aggregate. Error bars denote 95% bootstrap confidence intervals for the score aggregates.

5 Discussion

LatentGoalOps clears the benchmark-science checks we care about most: baseline ordering, a persistent oracle gap, a consistent shift penalty, and no easy shortcut on the two anchor tasks. Because malformed actions change the trajectory itself, strict score, native-only score, and native episode rate need to be reported together.

6 Limitations and Intended Use

- The benchmark is synthetic and intended for comparative evaluation, not operational advice or direct real-world deployment claims.
- The heldout split preserves schema and grader semantics, and P3 is treated as shorter planning breadth rather than as a load-bearing anchor.
- GPT-OSS 120B is API-served, and this version does not include human baselines or a full human-audited grader validation study.
- The adaptation metric uses thresholded detection rules; we expose and audit those choices in the release protocol, but smoother alternatives remain possible.

7 Conclusion

LatentGoalOps isolates hidden-objective adaptation under silent non-stationarity. On the tighter P1/P2 anchor slice and the broader release aggregate, the benchmark shows no easy shortcut on the anchor tasks, a persistent shift penalty, and a clear gap between current local models and the myopic-oracle reference, while also revealing that some sharp heldout failures are interface-mediated. That combination argues for evaluating latent policy quality and action-shell reliability separately.

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